

PES2UG122CS589

i) What are RNNs?

⇒ RNN are neural networks designed to work with sequential data by maintaining an internal state or "memory". Unlike state neural networks that process inputs independently, RNNs process data in sequence, with each step's output depending on previous computations.

Suhas.S.MWorking:-

1. They take an input sequence.
2. At each step, they
 - * Process current input
 - * Combine it with their internal state
 - * Update their state
 - * Produce an output.
3. Key applications include:
 - * Natural language processing
 - * Speech recognition
 - * Time series prediction.
 - * Music generation.

However, basic RNN have limitations with long sequences due to vanishing/exploding gradients. This led to improved variants like LSTM & GRU.

2. How RNN works for machine translation ⇒ Basic structure:

1. Encoder RNN:

- * Processes the input sentence in source language
- * Creates a fixed length context vector representing entire input.
- * Each word feeds into hidden state, accumulating meaning.

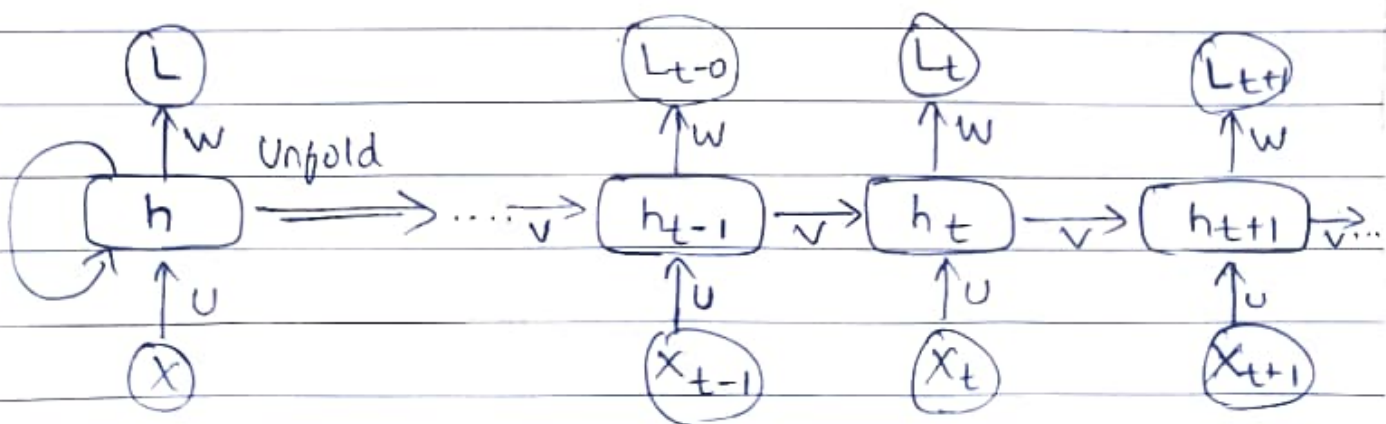
(Sahas.S.M)

2. Decoder RNN:

- * Takes the context vector as initial state.
- * Generates output sentence in target language.
- * Each step predicts next word based on previous predictions

Key improvements:

1. Attention mechanism
2. Bidirectional RNNs



3. End to end speech ~~reg.~~ recognition with recurrent neural networks

⇒ End to end speech recognition ~~to~~ using Recurrent neural networks (RNNs) represents a significant step forward from traditional automatic speech recognition (ASR) systems. Traditional ASR systems typically involve several stages, including feature ~~extra~~ extraction, acoustic modeling, language modeling & pronunciation modeling. These systems are often complex, requiring extensive domain-specific knowledge & significant engineering efforts.

(Suhas.S.M)

In contrast, end to end speech recognition systems streamline this process by attempting to map audio waveforms directly to text, bypassing intermediate steps. RNN based end to end ~~sa~~ systems are a popular approach for achieving this ~~the~~.

Training process :-

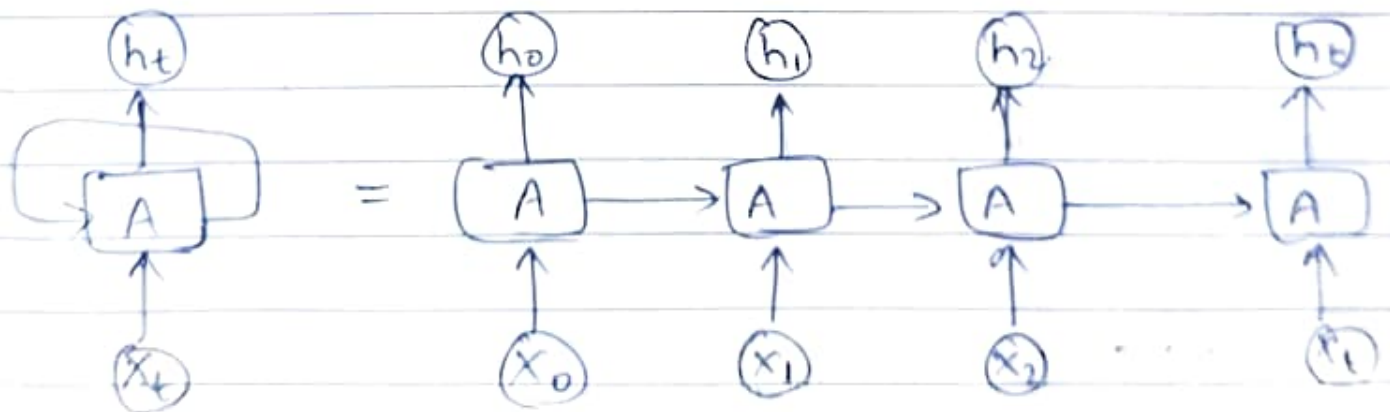
1. Data preprocessing
2. Model training
3. Evaluation metrics
4. Fine tuning & transfer learning.

4. Bi-directional RNN

A Bi-directional Recurrent neural network is a type of RNN that processes data in both forward & backward directions, allowing it to capture context from both past & future in a sequence. This sequence architecture is particularly beneficial for tasks like speech recognition, natural language processing, & other sequential data applications, where understanding the context from both directions can improve accuracy.

Applications:-

- * Speech recognition.
- * Natural language processing.
- * Machine translation.



A Bidirectional RNN has: Input layer, 2 RNN layers, Concatenation Layer & an output layer.

5. Deep (Bi-directional) RNN

→ A deep bi-directional recurrent neural network is an advanced form of bi-directional RNN that stacks multiple layers of bi-directional RNNs, resulting in a network that can capture complex patterns in data more effectively. Deep bi-RNNs combine the benefits of depth with bi-directionality, allowing them to learn highly expressive representations that are particularly valuable in complex sequential data tasks such as automatic speech recognition, language modeling & time series analysis.

(Suhas.S.M)

Architecture:

- * Input layer
- * Multiple bi-directional RNN layers
- * Output layers

Applications:-

1. Automatic Speech Recognition
2. Natural language processing
3. Time series analysis.

Deep bi-directional RNNs extend the capabilities of traditional RNNs by combining the power of multiple layers with bi-directional context.

6. Long short term Memory (LSTM)

=> LSTM is a specialized type of recurrent

neural network designed to address the limitations of traditional RNNs, particularly their difficulty in capturing long term dependencies due to issues like the vanishing gradient problem. LSTMs achieve this by introducing a memory cell & gating mechanisms that allow the network to maintain, update, or forget info over long sequences, making them ideal for tasks involving sequential data, such as time series prediction, language modeling, & speech recognition.

Applications:-

(Suhas S.M)

- * Speech recognition
- * Natural language processing
- * Time series forecasting
- * Video analysis

Challenges:-

- * Computational complexity
- * Sensitivity to long sequence
- * Prone to overfitting
- * Need for Tuning

7. Vanilla RNN vs LSTM.

⇒ Vanilla RNN & LSTMs are both types of Recurrent neural networks (RNNs) designed for sequential data processing. However, they differ significantly in how they handle information over long sequences, and each has its strength & limitations.

Vanilla RNN.

- * Architecture is simple, single hidden state

- * Ability to handle long term dependencies is limited.

- * Vanishing gradient problem is affected significantly

- * Computational cost is low

- * Training complexity is easier

- * Used for short sequences, simple tasks

- * No explicit memory

 PES UNIVERSITY mechanism.

LSTM.

- * Architecture has memory cell with gates

- * Ability to handle long term dependencies is better

- * It is less affected.

- * Computational cost is higher due to gates

- * Training complexity is harder

- * Used for long sequences, complex dependencies

- * Dedicated memory cell.